25901555

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACSTA200 - ETHICAL DIMENSIONS IN STATISTICAL MACHINE LEARNING THROUGH R/PYTHON

SCHEME OF VALUATION

Part A – MCQ - Answer all questions

Give 1 mark to each correct answer

1. d). Identify errors and ambiguities
2. b). Weaker the relationship
3. c). Systematic presentation of data in rows and columns
4. b). Populations are normally distributed
5. b). Chi-square test of independence
6. d). Categorical
7. Give 1 mark to any attempt
8. c). One dependent and one independent variable
9. c). Ordinal
10. c). Virtue ethics

Part B - Short Essay Questions - Answer any 5 questions

Give 6 marks to each correct answer

11. Arithmetic Mean - Explanation/ Definition & Formula - 4 Marks
Formula only – 2 Marks
At least one Example – 2 Marks

12. Ogives – Definition 3 marks
Types - Name and explanation - 3 marks
Types - Name only - 2 marks
Points for reference
ogive is a line graph that represents the cumulative frequency distribution of grouped data; two types of ogives- less than ogive and greater than or more than ogive

13. Correct explanation on Markov Chain Monte Carlo methods – 6 marks
Points for reference
MCMC are powerful Monte Carlo methods often used in Bayesian inference; used to generate sequences of dependent observations; can handle Complex and High-Dimensional Distributions; MCMC provides a full range of samples, allowing for the

estimation of credible intervals; MCMC is used to study how diseases spread; MCMC methods are applied to model stock prices and financial risks.

14. **Bayesian inference – Explanation - 2 marks**
    Prior distribution - Definition - 2 marks
    Posterior distribution – Definition - 2 marks
    Points for reference
    Bayesian inference - updates the probability as more data becomes available; incorporates prior knowledge; Bayesian Inference is applied in spam filtering, clinical trials, portfolio optimization, etc
    Prior - represents knowledge about an unknown parameter before observing the data; Informative Priors; Uninformative (Flat) Priors
    Posterior – is the refined belief; is the probability distribution of the parameters given the observed data
15. **Correct explanation on the principle of accountability and responsibility in data science – 6 Marks**
    Points for reference
    - obligation to take ownership of data-driven processes
    - answerable for their actions and the impacts
    - promote trust, fairness, etc.
16. **Correct explanation of the role of ethical frameworks in handling sensitive and personal data during analysis – 6 Marks**
    Points for reference
    - Promote Responsible Decision-Making
    - Protect Individual Rights
    - Reduce Bias and Unfairness
    - Ensure Transparency and Accountability
    - Maintain Data Integrity and Accuracy
    - Build Public Trust
17. **Describing all data types - 6 Marks**
    Naming the data types – 4 Marks
    For reference - Data Types in R
    - Numeric
    - Integer
    - Character
    - Logical
    - Complex

Part C - Essay Questions - Answer any 1 question

Give 10 marks to the correct answer

18. **Independence of attributes – Correct explanation 6 marks**

    Criterion for independence - Correct explanation 4 Marks

    Points for reference

    - Independence of attributes - does not exist any kind of relationship, we expect (i) the same proportion of A's in B's as in β's and (ii) same proportion of B's in A's as in α's
    - Criterion of independence - observed frequency equals the expected frequency AND/ OR the measures of association

19. **Working principle of the k-Nearest Neighbours (KNN) classification - Correct explanation 6 Marks**

    Strengths and limitations – At least 2 each – 4 Marks

    Points for reference

    Working Principle

    - classifies new observation based on the classes of the k closest observations in the training dataset
    - similar observations tend to belong to the same class
    - Steps
        - Select the value of k
        - Measure distance
        - Identify nearest neighbours
        - Assign class

    Strengths and limitations

    - Strengths - Simple to understand and implement; No explicit training phase; making it flexible; can model complex class boundaries; Effective when decision boundaries are irregular.
    - Limitations - Computationally expensive for large datasets; sensitive to noise and outliers, especially for small k; Performance depends on the choice of k and distance measure; affected by feature scaling, requiring normalization.